

USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations December 2025

20EC303 – ELECTRONIC DEVICES

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Only One full** question from **each Unit**.*

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) List the Steps involved in manufacturing Si Wafers. | 8 | L*2 | 1 | 1 |
| b) With neat cross-sectional diagrams, describe the fabrication steps of NPN Transistor. | 12 | L2 | 1 | 1 |
| 2. a) With a neat diagram, explain the process of vapor phase epitaxy. | 8 | L2 | 1 | 1 |
| b) With neat cross-sectional diagrams, describe the fabrication steps of nMOSFET. | 12 | L2 | 1 | 1 |

Unit – II

- | | | | | |
|---|----|----|---|---|
| 3. a) A uniform bar of n-type silicon of 2- μ m length has a voltage of 2 V applied across it. If $N_D = 10^{18}/\text{cm}^3$ and $\mu_n = 1350 \text{ cm}^2/\text{V}\cdot\text{s}$, find (a) the electron drift velocity, (b) the time it takes an electron to cross the 2- μ m length, (c) the drift current in the case that the silicon bar has a cross-sectional area of $0.25 \mu\text{m}^2$. | 06 | L3 | 2 | 1 |
| b) Write Short notes on Depletion Capacitance. | 04 | L2 | 2 | 1 |
| c) Using the Diodes 1N4148 having a $V_f = 0.5\text{V}$ Implement the Circuit blocks given in Figure Q3c if the input signal has a frequency of 10kHz. Choose Capacitor of $1\mu\text{F}$ | 10 | L3 | 2 | 1 |

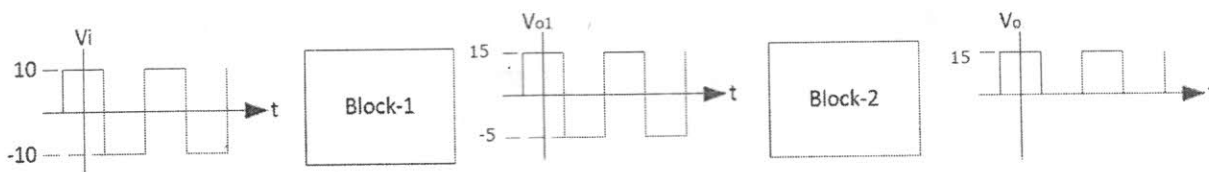


Figure Q3c

- | | | | | |
|--|----|----|---|---|
| 4. a) Consider a pn junction in equilibrium at room temperature ($T = 300 \text{ K}$) for which the doping concentrations are $N_A = 10^{18}/\text{cm}^3$ and $N_D = 10^{16}/\text{cm}^3$ and the cross-sectional area $A = 10^{-4} \text{ cm}^2$. Calculate V_0 , W , and Q_j . Use $n_i = 1.5 \times 10^{10}/\text{cm}^3$. | 06 | L3 | 2 | 1 |
| b) Write Short notes on Avalanche Breakdown. | 04 | L2 | 2 | 1 |
| c) With relevant sketches explain the working of Voltage Doubler. | 10 | L2 | 2 | 1 |

Unit – III

- | | | | | |
|---|----|----|---|---|
| 5. a) With relevant sketches explain the working of NPN Transistor in Active Mode. | 06 | L2 | 3 | 1 |
| b) In the circuit of Figure Q5b $I_{S1} = 2I_{S2} = 4 \times 10^{-16} \text{ A}$. Assuming $\beta_1 = \beta_2 = 100$ and $R_1 = 5\text{k}\Omega$, Calculate V_B such that $I_X = 1\text{mA}$ | | | | |

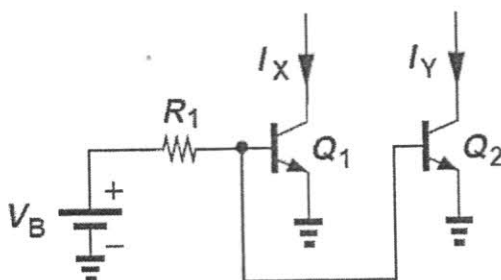


Figure Q5b

- c) Draw a neat sketch of high frequency model of a BJT and derive the expression for transit frequency.

08 L2 3 1

6. a) For an NPN transistor sketch of Minority carrier distribution and draw the output characteristic. With relevant sketches explain the variation in output characteristic due to early effect. Also draw the small signal model considering early effect.

06 L2 3 1

- b) For the circuit shown in Fig Q6b.,

- a) If $I_{S1} = I_{S2} = 5 \times 10^{-16} \text{ A}$, determine V_B such that $I_X = 1.2 \text{ mA}$.
b) What value of R_C places the transistors at the edge of active mode?

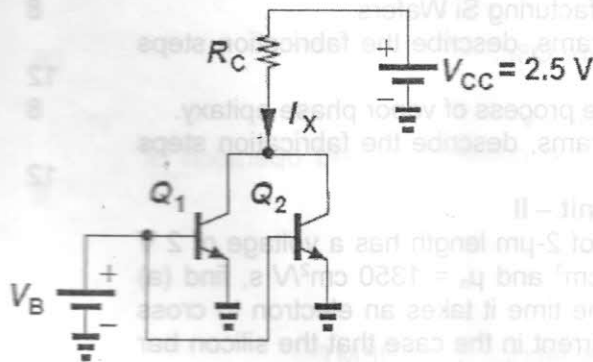


Fig Q6b.

06 L3 3 1

- c) Draw the small signal model for the circuit depicted in Figure Q6c(i) and Q6c(ii). Assume $V_A = \infty$.

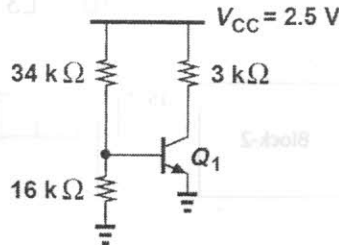


Figure Q6c(i)

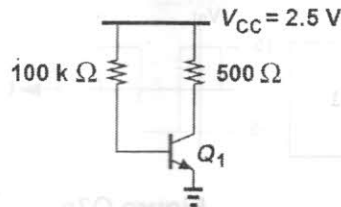


Figure Q6c(ii)

08 L2 3 1

Unit – IV

7. a) With respect to MOS transistors explain
(i) Transconductance
(ii) Channel Length Modulation
b) With neat diagrams, explain the formation of depletion region and channel in MOS transistor
c) An NMOS device operating in saturation must provide a transconductance of $1/(50 \Omega)$. Assuming $\mu_n C_{ox} = 200 \mu\text{A/V}^2$, $\lambda = 0$, and $V_{TH} = 0.4 \text{ V}$,
i. Determine W/L if $I_D = 0.5 \text{ mA}$.
ii. Determine W/L if $V_{GS} - V_{TH} = 0.5 \text{ V}$,
iii. Determine I_D if $V_{GS} - V_{TH} = 0.5 \text{ V}$

06 L2 4 1

08 L2 4 1

06 L3 4 1

8. a) With respect to MOS transistors define
(i) Body effect
(ii) Velocity saturation
b) Draw a neat diagram detailing the structure of an n-channel MOSFET. Explain its operation giving the plots of transfer and output characteristics

06 L2 4 1

08 L2 4 1

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- c) In the circuit of **Figure Q8c.**, $\mu_n C_{ox} = 200 \mu A/V^2$, $V_{th} = 0.4V$, $W/L = 20/0.18$ and $\lambda = 0$. What value of V_B places the transistor at the edge of saturation. Also determine the small signal parameters

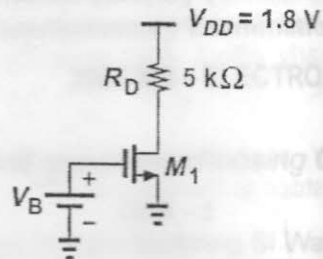


Figure Q8c.

06 L3 4 1

Unit – V

- | | | | | | | |
|-----|----|---|----|----|---|---|
| 9. | a) | With relevant sketches explain the construction and operation of Phototransistor. | 8 | L2 | 5 | 1 |
| | b) | Sketch Energy Band Diagrams for: | | | | |
| | | (i) MOS in Accumulation mode | 6 | L2 | 5 | 1 |
| | | (ii) MOS in depletion mode | 6 | L2 | 5 | 1 |
| | | (iii) MOS in inversion mode | | | | |
| | c) | With the neat diagram explain the fabrication of HEMT. | 12 | L2 | 5 | 1 |
| 10. | a) | With relevant sketches explain the fabrication of GaAs MESFET. | 8 | L2 | 5 | 1 |
| | b) | Explain the working of LED. | | | | |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome, PO* Program Outcome
